



DATABASE SYSTEM CONCEPTS & ARCHITECTURE

CHAPTER 2



DATA MODELS

- ★ **Data Model** → used to describe the structure of the database → helps to achieve data abstraction.

❖ Categories of Data Models

★ High-level or conceptual Data Model:

- Provides concepts that are close to the way many users perceive data.
- Use concepts such as entities, attributes and relationships.
- **Entities** → represent real world object or concept.
- **Attributes** → further describe an entity.
- **Relationships** → association among 2 or more entities.

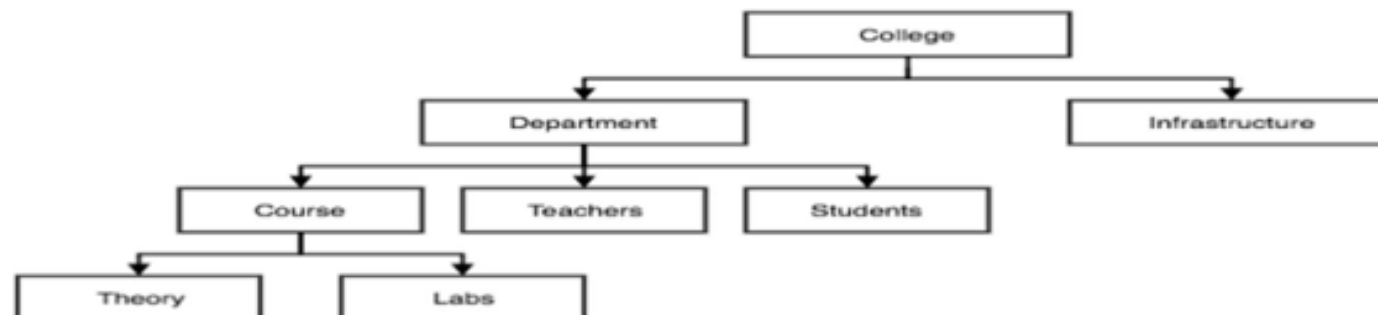
Representational data model

- It is used to represent the logical part of the database.
- One such popular model is the **Relational model**. In relational model, the database is organized in the form of tables/relations. We use 2D tables to store information inside the database.
- Technical term which is used for table in a relational database is **relation** and records are referred to as **tuples**.
- **Relational DB** is a collection of relations or 2D tables.

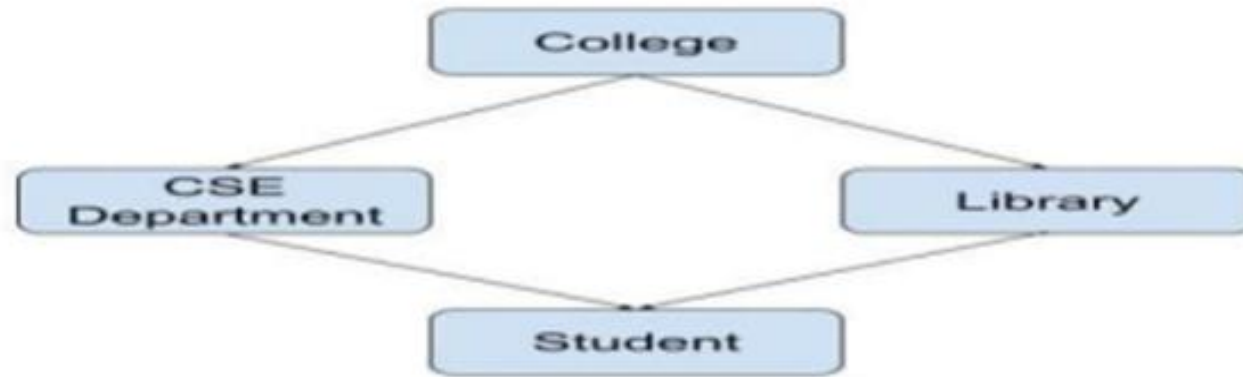
Representational data model

Other data models that are used to represent DB are

1. **Hierarchical model** - In a Hierarchical database, model data is organized in a tree-like structure. Data is represented using a parent-child relationship. In Hierarchical DBMS parent may have many children, but children have only one parent.



2. Network Model - The network database model allows each child to have multiple parents . In this model, entities are organized in a graph which can be accessed through several paths.



Network Model

❖ Categories of Data Models

★ Low-level or Physical Data Model:

- Describes **how data is stored** in the computer.
- **Access path** → structure for efficient searching of database records.

★ Representational (or implementation) Data Model:

- Represent data using **record structures** → record-based data models.



❖ Terminologies

- ★ **Database Schema:** Description of a database.
- ★ **Schema Diagram:** Displayed schema.

STUDENT

NAME	ROLL No.	BRANCH	ADDRESS
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- ★ **Schema Construct:** Each object within the schema.

Ex: STUDENT, COURSE, etc.

- ★ **Database State** (or **instance** or **snapshot**): The data in the database at a particular moment.

❖ Three-Schema Architecture

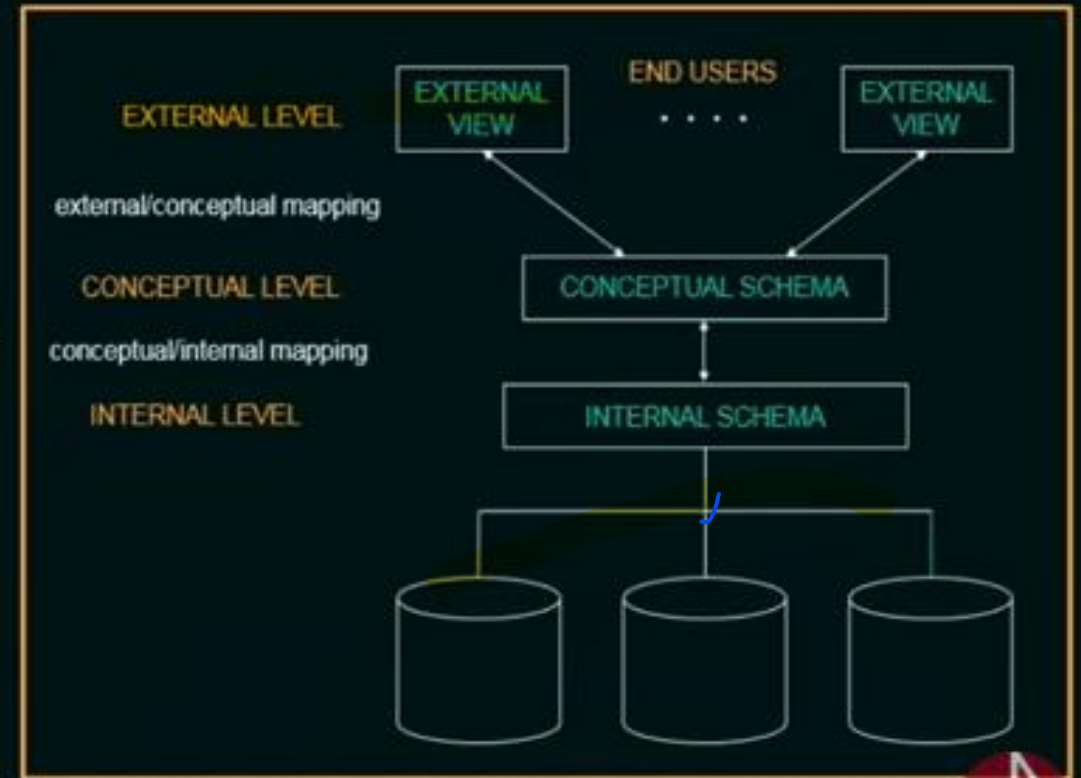
★ **Goal:** to separate the user applications and the physical database.

★ **3 Levels:**

● **Internal Level:**

→ Describes the physical storage structure of the database.

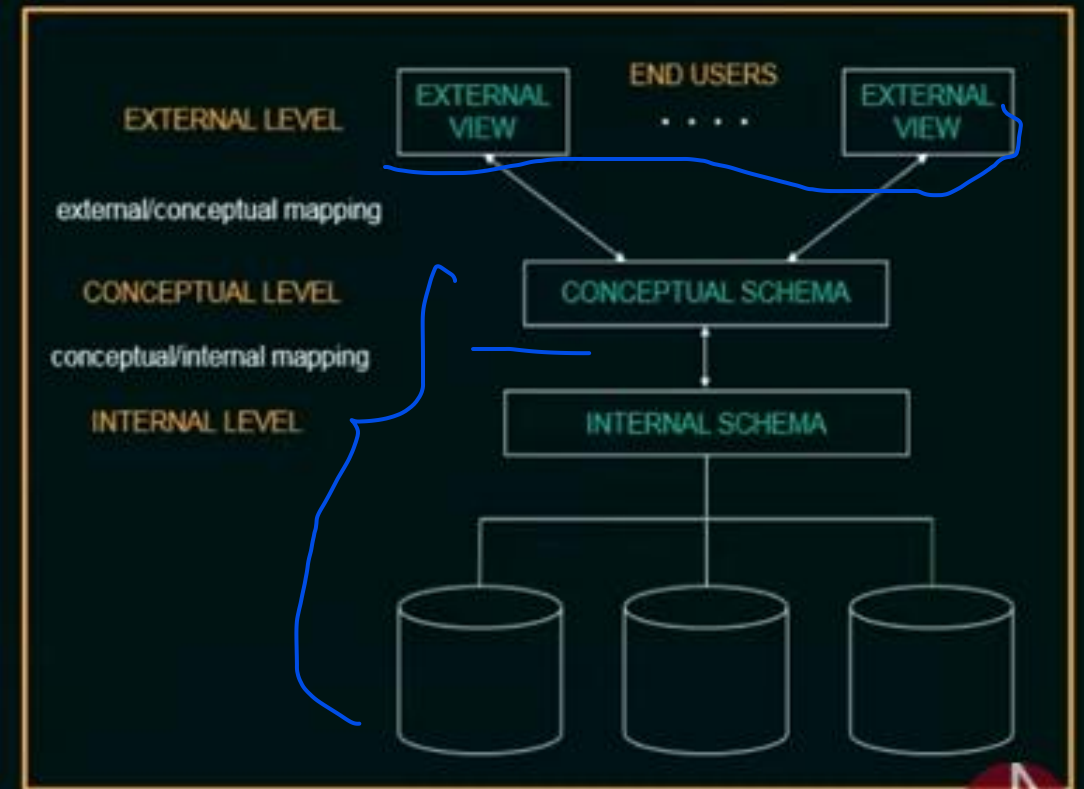
→ Describes complete details of data storage and access paths.



❖ Three-Schema Architecture

- Conceptual Level:

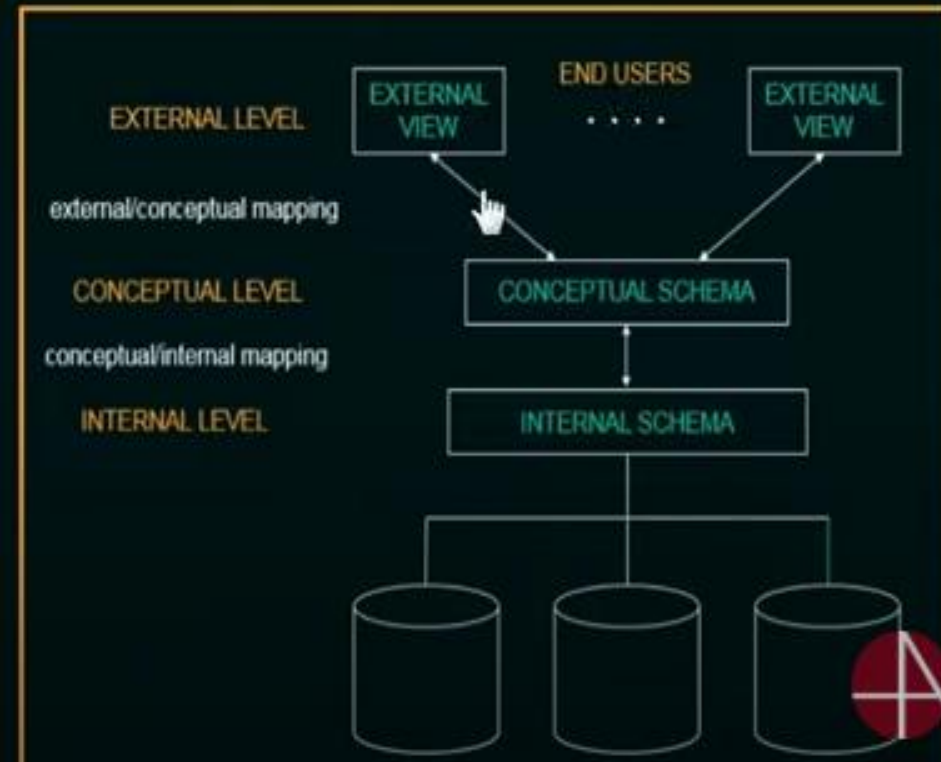
→ Hides the details of the physical storage structure and concentrates on describing entities, data types, relationships, constraints, etc.



❖ Three-Schema Architecture

- External Level:

→ Describes the part of the database that a user is interested in and hides the rest of the database from the user group.



DATA INDEPENDENCE

❖ Data Independence

✂ **Definition:** Capacity to change the schema at one level of a database system without having to change the schema at the next higher level.

★ Logical Data Independence:

- Ability to **modify** the **conceptual schema** without changing the external schemas or application programs.

★ Physical Data Independence:

- Ability to **modify** the **internal schema** without changing the conceptual schema.
- Changes may be needed → to **improve performance**.